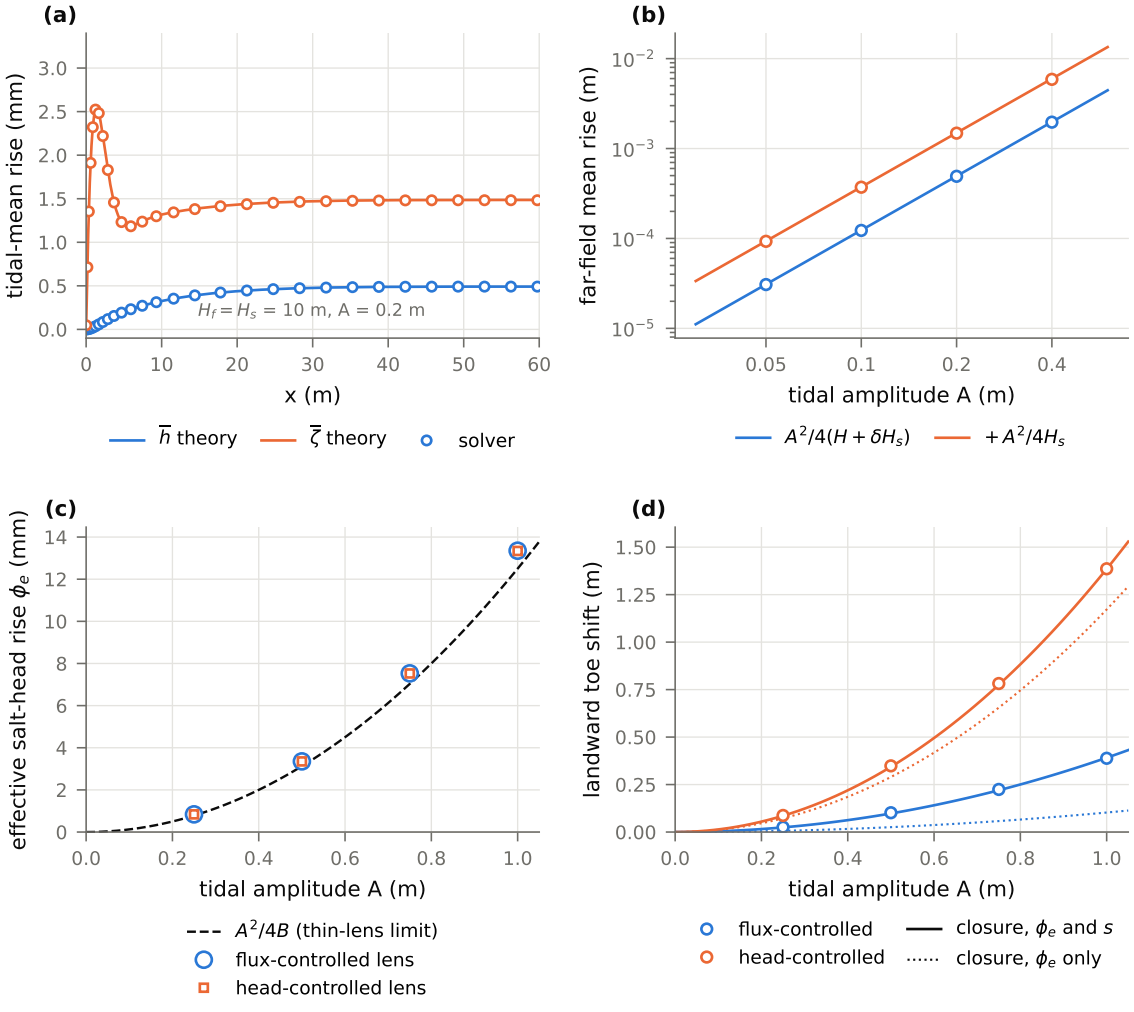
(a)
tidal-mean rise (mm)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
10
20
30
40
50
60
x (m)
$H_f = H_s$ = 10 m, A = 0.2 m
$\overline{h}$ theory
$\overline{\zeta}$ theory
solver
(b)
far-field mean rise (m)
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0.05
0.1
0.2
0.4
tidal amplitude A (m)
$A^2/4(H+\delta H_s)$
$+A^2/4H_s$
(c)
effective salt-head rise ϕ_e (mm)
14
12
10
8
6
4
2
0
0.0
0.2
0.4
0.6
0.8
1.0
tidal amplitude A (m)
$A^2/4B$ (thin-lens limit)
flux-controlled lens
head-controlled lens
(d)
landward toe shift (m)
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
tidal amplitude A (m)
flux-controlled
head-controlled
closure, ϕ_e and s
closure, ϕ_e only